

SELECT from WORLD Tutorial

Language:	English • 日本語 • 中文
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name	continent	area	population	gdp
Afghanistan	Asia	652230	25500100	20343000000
Albania	Europe	28748	2831741	12960000000
Algeria	Africa	2381741	37100000	188681000000
Andorra	Europe	468	78115	3712000000
Angola	Africa	1246700	20609294	100990000000
...				

In this tutorial you will use the SELECT command on the table world:

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1. Introduction

Read the notes about this table. Observe the result of running this SQL command to show the name, continent and population of all countries.

```
select name, continent, population from world;
```

Submit SQL

restore default

result

2. Large Countries

How to use WHERE to filter records. Show the name for the countries that have a population of at least 200 million. 200 million is 200000000, there are eight zeros.

```
select name from world where population>=200000000;
```

Submit SQL[restore default](#)

result

3. Per capita GDP

Give the name and the **per capita GDP** for those countries with a population of at least 200 million.

HELP:How to calculate per capita GDP

```
select name,gdp/population as per_capita_gdp from world where  
population>=200000000;
```

Submit SQL

restore default

result

0:00 / 2:19

4. South America In millions

Show the name and population in millions for the countries of the continent 'South America'. Divide the population by 1000000 to get population in millions.

```
select name, population/1000000 as new_population from world
where continent = "south america";
```

Submit SQL[restore default](#)

result

5. France, Germany, Italy

Show the name and population for France, Germany, Italy

```
select name, population from world where name in("france","germany","italy");
```

Submit SQL[restore default](#)

Correct answer

name	population
France	65906000
Germany	80716000
Italy	60782668

6. United

Show the countries which have a name that includes the word 'United'

```
select name from world where name like concat("united","", "%");
```

Submit SQL

restore default

Correct answer

name
United Arab Emirates
United Kingdom
United States

7. Two ways to be big

Two ways to be big: A country is **big** if it has an area of more than 3 million sq km or it has a population of more than 250 million.

Show the countries that are big by area or big by population. Show name, population and area.

```
select name, population, area from world where area>3000000 or population>250000000;
```

Submit SQL

restore default

result

8. One or the other (but not both)

Exclusive OR (XOR). Show the countries that are big by area (more than 3 million) or big by population (more than 250 million) but not both. Show name, population and area.

- Australia has a big area but a small population, it should be **included**.
- Indonesia has a big population but a small area, it should be **included**.
- China has a big population **and** big area, it should be **excluded**.
- United Kingdom has a small population and a small area, it should be **excluded**.

```
select name, population, area from world where area>3000000 xor population>250000000;
```

Submit SQL[restore default](#)

Correct answer

name	population	area
Australia	23545500	7692024
Brazil	202794000	8515767
Canada	35427524	9984670
Indonesia	252164800	1904569
Russia	146000000	17125242

9. Rounding

Show the name and population in millions and the GDP in billions for the countries of the continent 'South America'. Use the ROUND function to show the values to two decimal places.

For Americas show population in millions and GDP in billions both to 2 decimal places.

Millions and billions

Divide by 1000000 (6 zeros) for millions. Divide by 1000000000 (9 zeros) for billions.

Missing decimals

For some version of SQL the division of an integer by an integer will be an integer. One way to prevent this is to divide by a floating point number such as 1000000.0.

x

```
select name, round(population/1000000,2), round(gdp/1000000000,2)
from world where continent like 'South America';
```

Submit SQL[restore default](#)

Argentina	42.67	477.03
Bolivia	10.03	27.04
Brazil	202.79	2254.11
Chile	17.77	268.31
Colombia	47.66	369.81
Ecuador	15.77	87.50
Guyana	0.78	2.85
Paraguay	6.78	25.94
Peru	30.48	204.68
Saint Vincent and the Grenadines	0.11	0.69
Suriname	0.53	5.01
Uruguay	3.29	49.92
Venezuela	28.95	382.42

Play the SELECT card game (<https://sqlzoo.net/40289347/>)

1:14

SCORE: 5

```
SELECT card
FROM cards
WHERE number = 1
AND colour = 'red'
```

SELECT Card Game

1. Trillion dollar economies

Show the name and per-capita GDP for those countries with a GDP of at least one trillion (1000000000000; that is 12 zeros). Round this value to the nearest 1000.

Show per-capita GDP for the trillion dollar countries to the nearest \$1000.

```
select name, round(gdp/population,-3) as per_capita_gdp from world where gdp>=1000000000000;
```

Submit SQL

restore default

Canada	45000
China	6000
France	40000
Germany	42000
India	2000
Italy	33000
Japan	47000
Mexico	10000
Russia	14000
South Korea	22000
Spain	28000
United Kingdom	39000
United States	51000

1. Name and capital have the same length

Greece has capital Athens.

Each of the strings 'Greece', and 'Athens' has 6 characters.

Show the name and capital where the name and the capital have the same number of characters.

- You can use the LENGTH function to find the number of characters in a string

For Microsoft SQL Server the function LENGTH is LEN

```
select name, capital from world where length(name) = length(capital);
```

Submit SQL

restore default

Correct answer

name	capital
Algeria	Algiers
Angola	Luanda
Armenia	Yerevan
Botswana	Gaborone
Canada	Ottowa
Djibouti	Djibouti

Egypt	Cairo
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Variations

base

```
SELECT name, LENGTH(name), continent,
LENGTH(continent), capital, LENGTH(capital)
FROM world
WHERE name LIKE 'G%'
```

mssql

```
SELECT name, LEN(name), continent,
LEN(continent), capital, LEN(capital)
FROM world
WHERE name LIKE 'G%'
```

mysql

```
SELECT name, LENGTH(name), continent,
LENGTH(continent), capital, LENGTH(capital)
FROM world
WHERE name LIKE 'G%'
```

oracle

```
SELECT name, LENGTH(name), continent,
LENGTH(continent), capital, LENGTH(capital)
FROM world
WHERE name LIKE 'G%'
```

postgres

```
SELECT name, LENGTH(name), continent,
LENGTH(continent), capital, LENGTH(capital)
FROM world
WHERE name LIKE 'G%'
```

1. Matching name and capital

The capital of Sweden is Stockholm. Both words start with the letter 'S'.

Show the name and the capital where the first letters of each match. Don't include countries where the name and the capital are the same word.

- You can use the function LEFT to isolate the first character.
- You can use <> as the **NOT EQUALS** operator.

```
select name, capital from world where
substr(name,1,1) = substr(capital,1,1) and name <> capital;
```

Submit SQL

restore default

Correct answer

name	capital
Algeria	Algiers
Andorra	Andorra la Vella
Barbados	Bridgetown
Belize	Belmopan
Brazil	Brasília
Brunei	Bandar Seri Begawan
Burundi	Bujumbura



All the vowels

Equatorial Guinea and **Dominican Republic** have all of the vowels (a e i o u) in the name. They don't count because they have more than one word in the name.

Find the country that has all the vowels and no spaces in its name.

- You can use the phrase `name NOT LIKE '%a%'` to exclude characters from your results.
- The query shown misses countries like Bahamas and Belarus because they contain at least one 'a'

```
select name from world where name like "%a%" and name like "%e%" and name like "%i%" and name like "%o%" and name like "%u%" and name not like concat("%", " ", "%") ;
```

Submit SQL

restore default

Correct answer

name
Mozambique

What Next

- BBC QUIZ
- You can to continue practising the the same techniques and gain more experience of the basic skills on the Nobel table. The `WHERE` statement using the `nobel` table.
- You can learn about nested statements, these are instructive and entertaining, but not essential for beginners. Nested `SELECT` statements using the `world` table.

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The banner features a blue robot character on the left, a central text area, and a code editor on the right. The robot is a small, blue, friendly-looking character with large eyes and antennae. The central text area contains the title 'Master SQL', a subtitle 'Free, interactive lessons for total beginners', and a call to action 'Join 4M+ learners' with a 'START FREE' button. The code editor on the right shows a SQL query: 'SELECT name FROM learners WHERE course = 'SQL';'.

Master SQL

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```
query.sql

SELECT name
FROM learners
WHERE course = 'SQL';
```